Forecasting Demand and Reducing Food Waste in a Gas Station

DTSC 5082, Seminar in Research and Research Methodology (Section 401) - Fall 2024

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V T Tharun Sai
11617958
tharunsaivt@my.unt.edu

Siva Rohith Jampana
11689560
sivarohithjampana@my.unt.edu

Harshavardhan Solingaram
11617963
HarshavardhanSolingaram@my.unt.edu

Abstract

We are here to address the challenges in predicting daily sales for certain food items, namely Chicken Wings, Friedsnacks, Fried Burritos, and Other Food, sold at a gas station. Most of the time, gas stations face difficulties in optimizing inventories due to unpredictable customer demand, which leads to problems such as food waste and lost sales opportunities. This research makes use of historic sales data from the year 2017 through to 2024 and external drivers of demand, like holidays and price, in the development of predictive models based on machine learning. Advanced algorithms involving XGBoost and LSTM techniques have been employed in this case to make highly precise forecasts of sales trend tendencies. The exploratory data analysis included understanding patterns in sales higher at weekends and holidays, followed by finding significant lagged Sales variable correlations with current demand. The hypothesis testing also confirmed that the influence of holidays in sales behavior was valid. The performance of the different predictive models was compared, from which the LSTM model emerged as the best. The main conclusion is that predictive analytics may significantly reduce waste and enhance efficiency in inventory management related to food service operations at gas stations. This work serves as the basis for real-time forecasting systems and future optimization strategies to enhance decision-making processes.

The code, data, analysis, and results can be accessed on GitHub at: [https://github.com/TharunSaiVT/DTSC-5082]

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Keywords: Machine Learning Models, 1. Food Demand Forecasting, Inventory Management, Gas Station Optimization, XGBoost, LSTM (Long Short-Term Memory), Waste Reduction

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1 Introduction

Inventory management and demand forecasting are the two major challenges faced in the food industry, specifically those that operate through gas stations selling fast foods like Chicken Wings, fried snacks, Fried Burritos, and similar items. These problems arise simply due to the variation in customer demand for a particular item, which may depend on seasons, holidays, weekends, prices, etc. This is inconsistent with the demand pattern, which might cause overproduction, leading to food waste and financial loss, or underproduction, causing lost sales and dissatisfaction for customers. With increasing costs of food and a rising call for sustainability, making optimal food preparation decisions has become more relevant than ever. Traditional forecasting methods have, to a large degree, proven very limited in addressing complex nonlinear relationships and temporal dependencies in food sales data. Recently, the development of machine learning and advanced predictive models holds promise for solving some of these issues. Predictive models leverage historical sales data and external factors like holiday schedules and pricing trends that help enhance the accuracy and reliability of forecasts in demand.

The following study discusses the application of advanced statistical and machine learning models, XGBoost, and LSTM, in predicting daily sales for selected food items at a gas station.

The study shall adopt a data-driven approach to achieve better inventory management, reduction of food waste, and,

overall, more efficient operations. By addressing a real-world problem, this research contributes to the literature on predictive analytics while providing practical solutions for enhancing decision-making processes within the food service industry. .

1.1 Statement of the problem:

Gas station convenience stores face a critical challenge in demand forecasting for Cooked food. The current reliance on manual estimations or historical averages often leads to inaccurate demand predictions, which cause significant operational issues.

- Overproduction: Inaccurate demand forecasts can lead to overproduction, resulting in unnecessary food waste, financial losses, and environmental impact due to discarded products that do not sell in time.
- Underproduction: On the other hand, inaccurate forecasts may also cause underproduction, leading to stockouts, missed sales opportunities, and declining customer satisfaction due to unavailable products when customers need them.

This imbalance between supply and demand significantly affects both the bottom line and customer experience. Therefore, there is a pressing need to improve the accuracy of demand forecasting to optimize food inventory, minimize waste, and maximize sales.

2 Literature Review

Accurate demand forecasting is critical for managing food inventory, minimizing waste, and ensuring customer satisfaction. This section synthesizes recent advancements and methodologies in demand forecasting, focusing on machine learning, statistical analysis, and their applications in various domains, including convenience stores, restaurants, and catering services.

2.1 Challenges in Food Demand Forecasting

Demand forecasting in food supply chains is inherently challenging due to the perishable nature of food products, dynamic customer preferences, and external factors influencing demand. Panda and Mohanty [3] emphasize that traditional forecasting methods, such as time-series models, often fail to capture these complexities effectively. They propose a regressors-based analysis for modeling food demand and supply chains, demonstrating its superiority over conventional methods in predicting demand accurately.

Similarly, Rodrigues et al. [5] highlight the impact of inaccurate forecasts on food catering services, where underproduction leads to missed sales and customer dissatisfaction, while overproduction exacerbates food waste and financial losses. Their study employs machine learning models to address these challenges, demonstrating significant improvements in short-term demand prediction accuracy.

2.2 Machine Learning in Demand Forecasting

Machine learning has emerged as a transformative tool for demand forecasting. Rodrigues et al. [5] show that advanced machine learning models can effectively forecast demand for perishable goods, thereby reducing food waste and operational inefficiencies in catering services. The authors highlight the importance of feature selection and model tuning in achieving high predictive accuracy.

Tanizaki et al. [7] extend this application to restaurants, using machine learning and statistical analysis to forecast demand. Their study demonstrates that integrating machine learning with traditional statistical methods can capture complex demand patterns more effectively than standalone models.

Raschka [4] provides a comprehensive review of model evaluation, selection, and algorithm choice in machine learning. This work underscores the importance of choosing appropriate metrics and methodologies for evaluating forecasting models, ensuring they align with specific business objectives and constraints.

2.3 AI Applications in Convenience Stores

The integration of artificial intelligence (AI) in convenience stores has the potential to revolutionize demand forecasting and inventory management. According to Solink [6], AI video analytics can predict product demand by analyzing customer behavior and store traffic patterns, enabling more precise inventory adjustments and reducing waste.

Additionally, Algonomy [1] presents a four-step action plan leveraging AI for inventory planning in convenience stores. This approach not only minimizes stockouts and overstock issues but also enhances profitability by aligning inventory levels with real-time demand.

2.4 Perishable Food Demand Forecasting

Forecasting demand for perishable foods requires specialized approaches due to their limited shelf life. Nexocode [2] emphasizes the unique challenges posed by fresh food products, recommending tailored machine learning models that account for factors like spoilage rates and seasonal demand fluctuations. Their insights align with Panda and Mohanty [3]'s findings on the critical role of accurate demand predictions in reducing waste and improving supply chain efficiency.

2.5 Reducing Food Waste Through Forecasting

The environmental impact of food waste is a significant concern. Rodrigues et al. [5] and Nexocode [2] both emphasize the dual benefits of accurate demand forecasting: financial savings and environmental sustainability. By integrating short-term demand forecasting models, businesses can align production with actual demand, thus minimizing waste.

2.6 Statistical and Hybrid Approaches

While machine learning has gained prominence, traditional statistical methods still play a crucial role in demand forecasting. Tanizaki et al. [7] demonstrate the effectiveness of hybrid approaches that combine machine learning with statistical models. This combination leverages the strengths of both methodologies, offering robust and interpretable forecasting solutions.

2.7 Model Evaluation and Selection

Raschka [4] provides a framework for evaluating and selecting forecasting models, highlighting the importance of cross-validation, hyperparameter tuning, and algorithm selection. These practices are essential for ensuring that models generalize well to unseen data, a critical requirement for real-world applications.

2.8 Implications for Gas Station Convenience Stores

The insights from these studies are highly relevant for gas station convenience stores, where demand forecasting is essential for optimizing food inventory. The application of AI [1, 6], machine learning [5], and hybrid approaches [7] can address the unique challenges faced by these businesses. By adopting advanced forecasting models, gas station convenience stores can achieve significant reductions in food waste and improvements in customer satisfaction.

The literature demonstrates that demand forecasting for food products is a multifaceted problem requiring advanced methodologies. Machine learning and AI, combined with traditional statistical techniques, offer promising solutions for improving forecast accuracy, reducing waste, and enhancing operational efficiency. Future research could explore the integration of real-time data, such as weather conditions and traffic patterns, into forecasting models to further enhance their predictive capabilities.

3 Objectives of the Study

- The primary objective of this project is to develop an accurate short-term inventory prediction model specifically designed for cooked food items in gas station convenience stores. Given the rapid spoilage rate of cooked foods, the project focuses on creating a model that predicts demand on a short-term basis, considering historical sales data and external factors such as weather, local events, and holidays.
- To achieve this, the project aims to accomplish several key goals. Firstly, it seeks to develop a short-term demand forecasting model for cooked food items. Secondly, it aims to create a short-term inventory prediction system that aligns food preparation with predicted demand, thereby reducing overproduction and minimizing waste. Additionally, the project aims to reduce food waste and improve efficiency by forecasting demand accurately, leading to better resource utilization, lower costs, and improved environmental sustainability.
- Furthermore, the project seeks to optimize inventory levels for cooked foods, balancing the risk of stockouts with the need to avoid excess food preparation that may spoil. The effectiveness of the developed models will be evaluated using performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2.
- In addition to developing a short-term inventory prediction model, another key objective of this project is to identify and analyze the factors affecting food wastage in gas station convenience stores. This involves formulating hypotheses around these factors, such as the impact of weather conditions on hot food demand or the effect of holidays on sales, and then using statistical analysis and machine learning models to test these assumptions. The project also aims to evaluate the impact of external factors such as temperature, day of the week, and nearby events on food demand and waste.

4 Data Collection

The dataset for this project is sourced from the Point-of-Sale (POS) system of a gas station convenience store, spanning from 2017 to 2024.

4.1 Real Data (2021–2024)

- Includes actual sales data collected over the last few years
- Provides a reliable snapshot of the store's performance and sales patterns

4.2 Synthetic Data (2017–2020)

- Generated artificially for the purpose of model training
- Fills gaps in historical data and ensures a comprehensive dataset for training predictive models

4.3 Feature Variables

- Date: The date when the sales transaction occurred
- Is Holiday: A binary indicator showing whether the date is a public holiday or not
- Lag features: Previous week's sales for Chicken, FriedSnacks, FriedBurritos, OtherFood
- Moving average features: Smoothed demand patterns for Chicken, FriedSnacks, FriedBurritos, OtherFood

4.4 Price Variables:

- PriceChicken: The price of chicken products
- PriceFriedSnacks: The price of fried snacks
- PriceFriedBurritos: The price of fried burritos

```
In [40]:  df.columns

Out[40]: Index(['Chicken', 'FriedSnacks', 'FriedBurritos', 'Otherfood', 'Is Holiday',
         'Price_Chicken', 'Price_FriedSnacks', 'Price_FriedBurritos',
         'Price_Other_Food', 'Day_of_Week', 'Month', 'Weekday', 'Is_Holiday',
         'Is_Weekend', 'Lag_Chicken_Wings', 'Lag_Breakfast',
         'Lag_Fried_Burritos', 'Lag_Other_Food', 'MA_Chicken_Wings',
         'MA_Breakfast', 'MA_Fried_Burritos', 'MA_Other_Food'],
        dtype='object')
```

Figure 1. Variables of the Dataset

- PriceOtherFood: The price of other food items sold

4.5 Target Variables (Sales Data)

- Chicken: Sales quantity of chicken products sold on a given day
- FriedSnacks: Sales quantity of fried snacks
- FriedBurritos: Sales quantity of fried burritos
- OtherFood: Sales quantity of other food items

This data collection captures the relationship between product pricing, holidays, and food sales, serving as the foundation for further analysis and predictive modeling.

```
In [117]:  # Check the data types
           print(df.dtypes)

Chicken                int64
FriedSnacks            int64
FriedBurritos          int64
Otherfood              int64
Is Holiday              bool
Price_Chicken         float64
Price_FriedSnacks     float64
Price_FriedBurritos   float64
Price_Other_Food      float64
Day_of_Week             int32
Month                   int32
Weekday                 int32
Is_Holiday               bool
Is_Weekend              int32
Lag_Chicken_Wings     float64
Lag_Breakfast         float64
Lag_Fried_Burritos    float64
Lag_Other_Food        float64
MA_Chicken_Wings      float64
MA_Breakfast          float64
MA_Fried_Burritos     float64
MA_Other_Food         float64
dtype: object
```

Figure 2. Data types of the Variables

5 Exploratory data analysis (EDA) and Hypotheses for the Study

Overall Observations:

- The dataset appears to be relatively large, with 2500 data points.
- The data is distributed across different periods, with months and days of the week represented.
- There is a clear distinction between weekdays and weekends, as well as holidays and non-holidays.
- Impute missing values with median or mean.
- Use time series-specific methods like forward filling (if missing values are due to small gaps).

Time-Related Factors(fig 3):

- Weekday sales are higher than weekend sales.

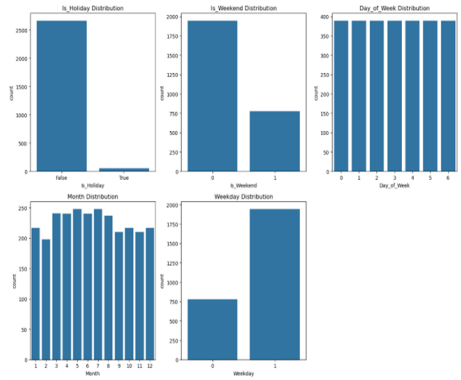

Figure 3. Time-Related Factors

- Sales vary across months, possibly due to seasonal factors.
- Holidays have a significant impact on sales.

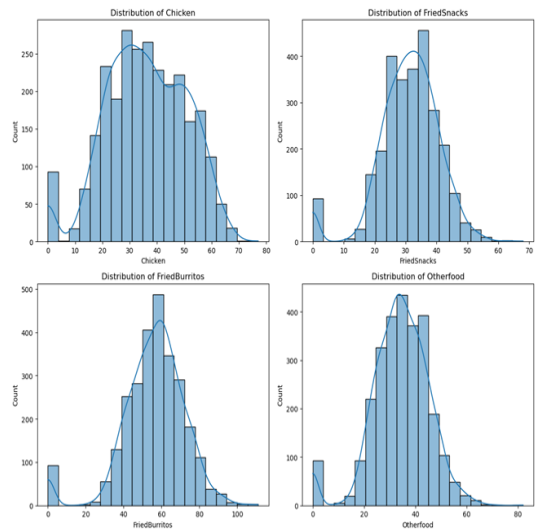

Figure 4. Sales Distribution

Sales Distribution(fig 4):

- Chicken: Right-skewed distribution, suggesting occasional high-demand periods.
- FriedSnacks, FriedBurritos, OtherFood: Approximately normal distributions, indicating stable sales patterns.

Time series of Food quantities(fig 5):

- Varied Demand Patterns: The time series plot reveals distinct demand patterns for each food item. Chicken exhibits periods of high demand, while FriedSnacks, FriedBurritos, and OtherFood show more stable and consistent sales.
- Potential Seasonal Trends: The plot suggests potential seasonal variations in demand, with certain periods experiencing higher or lower sales for specific food items.

Figure 5. Time series of Food quantities

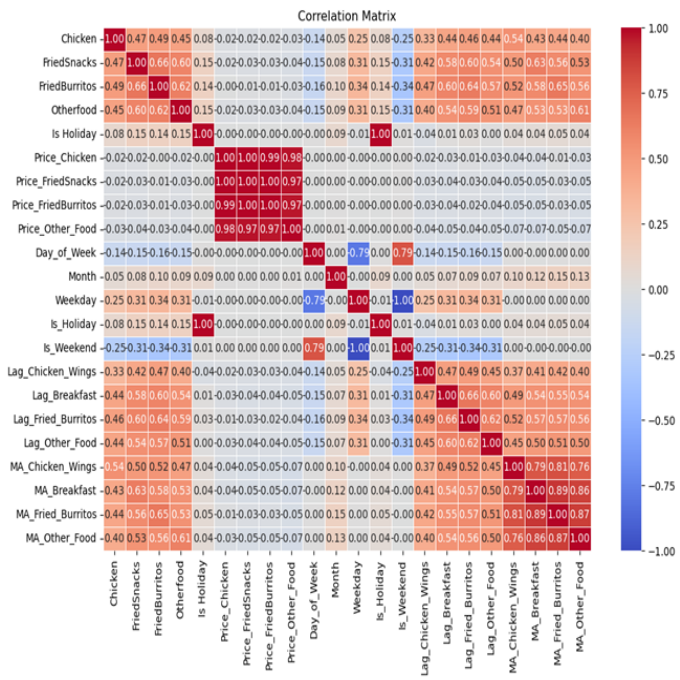

Figure 6. Correlation graph

Correlations and Feature relationships(fig 6):

- Strong positive correlations exist between the sales of different food items, indicating a joint impact on demand.
- Some negative correlations between food items and their lag features suggest potential demand shifts.
- Time-related features have weak correlations with food sales, implying a less significant direct impact.

6 Experiment and Data Analysis

6.1 Model Development:

To predict food sales at a gas station, two different high-end modeling techniques are used in the paper: XGBoost and LSTM. These two models are used because of their good efficiency in dealing with time series, non-linear relationships, and temporal dependencies. XGBoost stands for Extreme Gradient Boosting; it was employed here for being robust in handling multi-feature data sets, whereby interaction of variables such as holidays, price, and day of the week play a significant role. Finally, the deep learning models are LSTM networks, as the latter have the capability for retaining sequential information over long time intervals; thus, they are considered apt for multi-output forecasts. The training for these models will be from historical sales from 2017 to 2024 and use feature engineering: lagged sales, moving averages, holiday 'on' flag, and pricing trend. The proposed ensemble models leverage the breadth of the model space for analyzing simple to complex patterns in data. Rigorous hyperparameter tuning along with cross-validation will ensure optimized performance.

Figure 7. Flow chart of Data preprocessing

6.2 Model Evaluation:

To evaluate the effectiveness of the predictive models, we conducted an extensive comparison between their outputs and actual sales data. Performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) were used to quantify the accuracy of each model. The models were tested across multiple scenarios, such as holidays, weekends, and regular weekdays, to assess their ability to generalize across different sales patterns. Comparisons have shown that, for multi-output forecasts, especially with high variability in

sales, LSTM always outperforms XGBoost. The ability of the LSTM model to learn long-term dependencies and temporal dynamics proved crucial for the correct prediction of sales spikes during holidays and special events. XGBoost demonstrated resistance to non-linear relationships but needed extensive feature engineering, which only then allowed it to perform as well as LSTM.

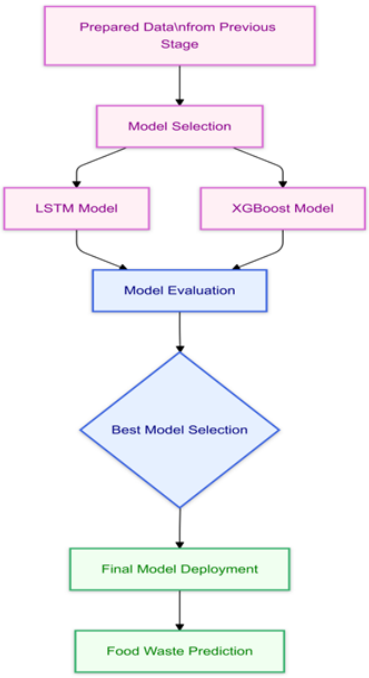

Figure 8. Model Evaluation

6.3 Data Analysis Process:

The data analysis process involved several steps:

- Preprocessing: The sales dataset collected was preprocessed to be analysis-ready. This included filling in missing values using interpolation techniques, removing extreme outliers that would distort model predictions, and standardizing the data format. For instance, dates were transformed into their respective components of year, month, day, and day-of-week for better feature representation. Also, holiday indicators were encoded so that they could be integrated into the modeling framework.
- Feature Engineering: Given that temporal factors are crucial for sales prediction, some lagged variables were created, such as sales from the previous day or week, moving averages, and seasonal indices. These features captured the historical dependencies and cyclical patterns that are crucial for the accuracy of forecasts. Furthermore, pricing adjustments and promotional events were added as categorical and continuous variables to quantify their impact on demand.
- Model Training and Evaluation: Every model was trained and tested on different splits to avoid overfitting. In the case of LSTM, sequence structures were prepared wherein every input sequence represented historical sales data over a certain period of time, and the output was the sales for the next period. The hyperparameter tuning for XGBoost and LSTM was done in terms of learning rate, batch size, and tree depth.
- Evaluation Metrics: The performance of the models was evaluated using MAE and RMSE, since they give a clear view of the prediction errors. Besides, MAPE was used to measure the percentage error, which can be interpreted in real-world terms. These metrics were computed for all target variables individually and collectively to measure overall performance.
- Experiment Design: Experiments were designed in a structured manner to study the effects different configurations have on model accuracy. For example, the role of lagged features was analyzed and its impact on model performance was measured. Similarly, the impact of external variables like weather and pricing was tested. Iterative experimentation ensured continuous improvement in the predictive models, with LSTM consistently emerging as the most robust.

	Model	RMSE	MAE	MAPE
0	XGBoost	8.614315	6.924497	30.374837
1	LSTM	9.410982	7.572233	22.286891

Figure 9. Evaluation Metrics

6.4 Parameter Settings:

For XGBoost, we have performed the tuning of parameters for the trade-off between predictive performance and computational efficiency. It was trained with 500 boosting rounds to allow it to iteratively refine predictions, while the maximum tree depth was set to 6 to capture sufficient complexity without overfitting. The learning rate was 0.1 for gradual updates toward stable optimization. For improving the generalization of the model, we performed sub-sampling with 80 percentage of training data for every tree and also sampled 80 percentage features per split. A child minimum weight of 1 and gamma value of 0 avoided any un-necessary splits by this model considering only major patterns. The regression used the reg:squarederror objective function in order to accurately minimize error. Controlled random state to reproduce the result. In the LSTM model, we designed a network architecture suitable for time-series forecasting. The length of the input sequence was set to 30 days to obtain temporal dependencies, while it had two stacked LSTM layers comprising 50 hidden units each in order to model complex patterns. A dropout rate of 0.2 prevents overfitting by randomly deactivating neurons during training. The Adam optimizer with adaptive

learning was used in conjunction with the Mean Squared Error loss function for efficient weight updates. Training was performed for 50 epochs with a batch size of 32, ensuring balanced computational load and convergence. A validation split of 20% allowed for monitoring overfitting, and a learning rate scheduler dynamically reduced the learning rate if validation loss plateaus, ensuring fine-tuned optimization. These configurations provided a substantial basis for the accurate prediction of multi-output sales.

7 Data Visualisation and Results

To achieve this, we conducted an in-depth analysis of predictions compared to actual data trends, providing insights into the efficiency and applicability of the models. The application of XGBoost and LSTM models to forecast food demand at a gas station has demonstrated promising results. The visualizations clearly show that while both models are effective in predicting demand, LSTM outperforms XGBoost in terms of accuracy and consistency, particularly for capturing time-dependent patterns. The performance metrics support this, with LSTM yielding lower error rates, making it a more reliable choice for forecasting demand over time.

In terms of forecasting accuracy, metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were employed. Across all food items, the results indicated that LSTM achieved lower error rates, demonstrating its capability to generalize well on unseen data. These insights enable stakeholders to identify key drivers of demand, enhancing decision-making processes.

Here are the forecasting results using LSTM model in which We predicted sales quantity for next 7 days.

Date	Chicken	FriedSnacks	FriedBurritos	Otherfood
2024-08-21	40.107643	31.758524	53.641918	34.603230
2024-08-22	55.000347	40.643997	58.810070	37.362503
2024-08-23	46.179920	43.919964	51.399410	42.341339
2024-08-24	49.757332	49.288071	67.514191	50.611992
2024-08-25	35.226627	45.254295	50.636883	31.508507
2024-08-26	37.161419	45.969299	51.055069	30.970245
2024-08-27	37.161419	45.969299	51.055069	30.970245

Figure 10. Forecast for next 7 days

The results emphasize the potential of data-driven forecasting models in operational efficiency. However, challenges related to noisy data, outliers, and abrupt demand changes during special events were observed. To address these, enhanced preprocessing techniques and the integration of external factors, such as weather patterns and economic indicators, could be explored in future work.

8 Conclusion and Limitations

The present study demonstrates the suitability of predictive models like XGBoost and LSTM for operational challenges in a gas station environment. The project successfully forecasted the demand for food items, thus providing actionable

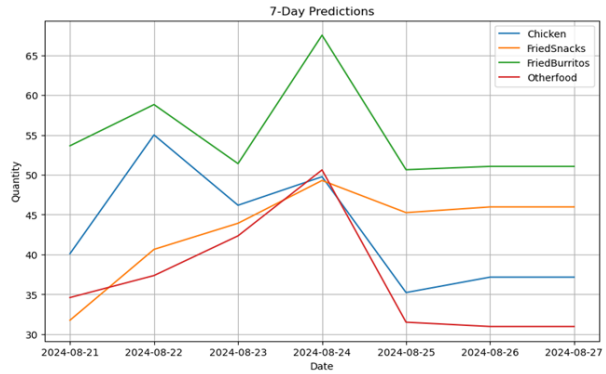

Figure 11. Graph for forecasting demand for next 7 days

insights into inventory management and waste reduction. Both models showcased promising results, but LSTM outperformed XGBoost in capturing temporal dependencies and delivered more accurate predictions. The visualizations provided further clarity on model performance and feature significance, underlining the relevance of contextual factors such as holidays and special events in driving demand.

```
Model Comparison:
XGBoost RMSE (per target): [11.90588538  5.89057083  9.43858935  7.22221446]
LSTM RMSE (per target): [12.3740534   6.82773713 10.47337949  7.968758  ]
XGBoost is the best model.
```

Figure 12. Best Model for prediction

Despite the encouraging results of some problems and limitations, data quality played a key role, where noise and inconsistencies in the historical sales data directly impacted model performance. Secondly, most models struggle to adapt to abrupt changes in demand triggered by unforeseen factors; thus, adaptive or hybrid approaches might be necessary in further implementations. This was especially regarding computational complexity, especially for LSTM, which requires efficient tuning of hyperparameters and effective resource management. This research underlines the potential of advanced machine learning models in smoothing operations and reducing wastage within retail environments. Going forward, the inclusion of more external features, refining preprocessing techniques, and the use of ensemble methods could lead to further improvements in prediction accuracy. This research lays the foundation for using predictive analytics in retail and presents a framework for optimizing inventory and resource management practices in similar contexts.

9 Future Work

Future research could improve the accuracy of forecasts by including external factors such as real-time weather, economic indicators, and traffic flow. Using advanced models like Transformers, could give better predictions. Adaptive

learning frameworks may allow models to learn from recent sales trends and anomalies, thereby improving resilience against changes in demand. Real-time monitoring systems could further optimize decision-making. Addressing domain-specific challenges, such as data imbalance and high variability during holidays, and democratizing access through user-friendly cloud-based tools could empower small retailers in better inventory management and waste reduction.

10 Author Contributions

1. Tharun Sai V T:
Tharun Sai was in charge of assigning tasks to the team members. he worked on the overall information and history of the idea/paper. In addition, he assisted in understanding the models and evaluating the models chosen for use in the project.

2. Harshavardhan Solingaram:
Harshavardhan was in charge of gathering information about datasets and fields that needed to be addressed to meet the paper's objectives. In addition, he assists in understanding the models that will work well with the specific data being worked on.

3. Siva Rohith Jampana:
Siva Rohith was in charge of gathering and comprehending relevant work on the paper. Also, design the best possible model architecture that can assist in the attempt to solve the current problem.

We have documented all that is required in the process of solving the problem statement of the paper, keeping in mind the contribution of every member, coming up with this proposal to improvise further in order to implement our ideas.

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